

# Shaibal Saha

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[!\[\]\(919a2cb85b99741a73c0c31a427236a8\_img.jpg\) LinkedIn](#) | [!\[\]\(c9cd5a1c35167a83f09a35036fe5dcbd\_img.jpg\) GitHub](#) | [!\[\]\(ae1936640fabdea8c18f922ca69733fe\_img.jpg\) Google Scholar](#)

Auburn Hills, Michigan - 48326, USA

## EDUCATION

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- **Oakland University** *September 2022 - August 2027 (Expected)*  
*PhD Student, Department of Computer Science and Informatics* Rochester Hills, USA  
*CGPA: 4.00 / 4.00*
- **Oakland University** *September 2022 - December 2024*  
*Masters of Science, Department of Computer Science and Informatics* Rochester Hills, USA  
*CGPA: 4.00 / 4.00*
- **North South University** *January 2014-August 2018*  
*Bachelor of Science in Computer Science and Engineering* Dhaka, Bangladesh  
*CGPA: 3.44 / 4.00*

## EXPERIENCE

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- **Oakland University** *September 2022 - April 2025*  
*Graduate Research Assistant* Rochester Hills, USA
  - Conduct research on optimization techniques for deep neural networks in Edge environments to enhance performance and efficiency.
  - Implement and analyze new model compression techniques, including pruning and quantization, for transformers and hybrid models for generalized application/domain-specific applications.
  - Conduct research on medical imaging task optimization techniques for heterogeneous platforms such as FPGA, Jetson Nano, and Nvidia AGX Orin.
  - Study multi-modal detection models for Autonomous Driving Systems, focusing on hardware/software co-design for optimized deployment.
  - **Frequently used tools:** PyTorch, TensorRT, Vitis-AI, Vaitrace, Monai, Numpy, Pandas, CUDA
- **Code Aristos** *January 2019 - June 2022*  
*Software Developer* Dhaka, Bangladesh
  - Started as an intern (Jan 2019 - Apr 2019) and transitioned to a full-time role due to strong performance.
  - Worked on Hospital Management ERP for large-scale data handling for both web and mobile apps.
  - Provided real-time client support to bug fixes for live systems all across the world.
  - Designed and implemented robust backend functions for multi-level user access and customization based on client needs.
  - **Frequently used tools:** Codeigniter, Django, Laravel, MongoDB, MySQL, Azure, Apache

## RESEARCH INTEREST

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Deep learning, Edge Computing, Computer vision, Cache in LLM, Model Optimizations, Connected Health, Autonomous Driving

## PUBLICATIONS

C=CONFERENCE, J=JOURNAL

- [J.1] **Shaibal Saha**, Lanyu Xu. Vision Transformers on the Edge: A Comprehensive Survey of Model Compression and Acceleration Strategies. arXiv, DOI: <https://doi.org/10.48550/arXiv.2503.02891> (Accepted in NeuroComputing).
- [C.1] Yunge Li, **Shaibal Saha**, Lanyu Xu (2024). The Architectural Implications of Multi-modal Detection Models for Autonomous Driving Systems. In *IEEE International Conference on Mobility: Operations, Services, and Technologies 2024 (MOST 2024)*, DOI: 10.1109/MOST60774.2024.00030o
- [C.2] Ronald Tudu, **Shaibal Saha**, Prasun Nandy Pritam and Rajesh Palit (2018). Performance Analysis of Supervised Machine Learning Approaches for Bengali Text Categorization. In *2018 5th IEEE Asia-Pacific World Congress on Computer Science and Engineering (APWC on CSE)*, DOI: 10.1109/APWC on CSE.2018.00043

## MANUSCRIPTS UNDER REVIEW

C=CONFERENCE, J=JOURNAL, B=BOOK CHAPTER

- [B.1] **Shaibal Saha**, Qiren Wang, Lichen Xia, Yongtao Yao. "Chapter 4 Towards Edge Intelligence." In *Edge Computing: Systems and Applications*, edited by Lanyu Xu, Weisong Shi
- [C.1] **Shaibal Saha**, Lanyu Xu. Paper on post-training quantization for hybrid models on edge devices Under review

## SKILLS

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- **Programming Languages:** [Expert] Python, PHP, VHDL  
[Moderate] C, C++, Javascript
- **Tools:** Jupyter, PyTorch, TensorFlow, TensorRT, Pandas, Monai, NumPy, Scikit-learn, Vaitrace
- **Hardware used:** GPU, Xilinx ZCU104 MPSoC, Jetson Nano, NVIDIA Jetson AGX Xavier
- **Other Technologies:** Docker, Vitis-AI, Intel Quartus, Vivado Design suite, Microsoft Azure, Apache, AWS, Huggingface
- **Web Technologies:** Django, Codeigniter, Laravel
- **Database Systems:** MySQL, NoSQL(MongoDB), Oracle, RedisStack, Redis
- **Logging & monitoring:** Drawio, Jira, Github, Git, Gitlab, ELK Stack, Huggingface
- **Completed Courses:** Embedded System Design Using FPGAs, Introduction to EdgeAI, Computer Architecture, Advanced Data Structure and Algorithms, Cloud Computing, Database System, Discrete methods I, Component-based software engineering.
- **Languages:** English (Fluent), Bangla (Native), Hindi (Professional)

## PROFESSIONAL SERVICES

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- Reviewer of IEEE transactions on emerging topics in computing [2025-Present]
- Reviewer of Journal of Ambient Intelligence and Smart Environment [2023-Present]
- Reviewer of 4th Workshop on Multi-lingual Representation Learning [2024]

## HONORS AND AWARDS

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- NSF Student Travel Award for attending IEEE/ACM SEC 2023
- Bangladesh-Sweden Trust Fund (BSTF)- Travel Grant 2023
- NSF Student Travel Award for attending IEEE/ACM CHASE 2023
- Oakland University Provost Graduate Student Research Award grant - Spring 2023
- Selected for university special scholarship at the undergraduate for maintaining good result

## LEADERSHIP EXPERIENCE & PROFESSIONAL MEMBERSHIPS

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- **ACM Student Chapter**, President *September 2024 - Present*
- **ACM Student Chapter**, Treasurer *September 2023 - August 2024*

## REFERENCES

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1. **Lanyu Xu, Ph.D**  
Assistant Professor, Computer Science and Engineering Department  
Oakland University  
Email: lxu@oakland.edu  
Phone: (248)370-4079  
*Relationship: PhD Advisor*
2. **Debatosh Debnath, Ph.D**  
Associate Professor, Computer Science and Engineering Department  
Oakland University  
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*Relationship: Course Instructor and Organization Advisor*